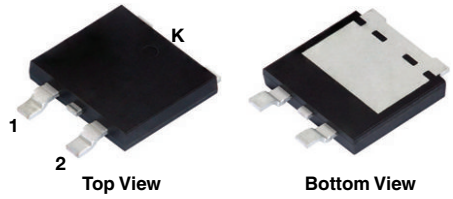
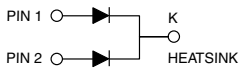
# Dual High-Voltage TMBS® (Trench MOS Barrier Schottky) Rectifier

**eSMP® Series**
**SMPD (TO-263AC)**




Top View Bottom View

**V20D202C**




## FEATURES

- Trench MOS Schottky technology generation 2
- Very low profile - typical height of 1.7 mm
- Ideal for automated placement
- Low forward voltage drop, low power losses
- High efficiency operation
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- AEC-Q101 qualified available:
  - Automotive ordering code: base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE GRADE Available

RoHS COMPLIANT

HALOGEN FREE

## LINKS TO ADDITIONAL RESOURCES

3D Models

## TYPICAL APPLICATIONS

For use in high frequency DC/DC converters, switching power supplies, freewheeling diodes, OR-ing diode, and reverse battery protection.

## MECHANICAL DATA

**Case:** SMPD (TO-263AC)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-M3 - halogen-free, RoHS-compliant
Base P/NHM3_X - halogen-free, RoHS-compliant, and AEC-Q101 qualified
(X_denotes revision code e.g. A, B, ...)

**Terminals:** matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
M3 and HM3 suffix meets JESD 201 class 2 whisker test

**Polarity:** As marked

| PRIMARY CHARACTERISTICS | |
|---|---|
| $I_{F(AV)}$ | 2 x 10 A |
| $V_{RRM}$ | 200 V |
| $I_{FSM}$ | 150 A |
| $V_F$ at $I_F$ = 10.0 A ($T_A$ = 125 °C) | 0.68 V |
| $T_J$ max. | 175 °C |
| Package | SMPD (TO-263AC) |
| Circuit configuration | Common cathode |

**MAXIMUM RATINGS** ($T_A$ = 25 °C unless otherwise noted)

| PARAMETER | | SYMBOL | V20D202C | UNIT |
|---|---|---|---|---|
| Maximum repetitive peak reverse voltage | | $V_{RRM}$ | 200 | V |
| Maximum average forward rectified current (fig. 1) | per device | $I_{F(AV)}$ | 20 | A |
| | per diode | | 10 | |
| Maximum DC reverse voltage | | $V_{DC}$ | 160 | V |
| Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load | | $I_{FSM}$ | 150 | A |
| Voltage rate of change (rated $V_R$) | | dV/dt | 10 000 | V/µs |
| Operating junction and storage temperature range | | $T_J$, $T_{STG}$ | -40 to +175 | °C |


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| ELECTRICAL CHARACTERISTICS ($T_A$ = 25 °C unless otherwise noted) | | | | | | |
|---|---|---|---|---|---|---|
| PARAMETER | TEST CONDITIONS | | SYMBOL | TYP. | MAX. | UNIT |
| Instantaneous forward voltage per diode | $I_F$ = 5 A | $T_A$ = 25 °C | $V_F$ (1) | 0.75 | - | V |
| | $I_F$ = 10 A | | | 0.83 | 0.9 | |
| | $I_F$ = 5 A | $T_A$ = 125 °C | | 0.6 | - | |
| | $I_F$ = 10 A | | | 0.68 | 0.76 | |
| Reverse current at rated $V_R$ per diode | $V_R$ = 160 V | $T_A$ = 25 °C | $I_R$ (2) | 0.8 | - | µA |
| | | $T_A$ = 125 °C | | 1 | - | mA |
| | $V_R$ = 200 V | $T_A$ = 25 °C | | - | 150 | µA |
| | | $T_A$ = 125 °C | | 2.5 | 10 | mA |

**Notes**

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 5 ms

| THERMAL CHARACTERISTICS ($T_A$ = 25 °C unless otherwise noted) | | | | |
|---|---|---|---|---|
| PARAMETER | | SYMBOL | V20D202C | UNIT |
| Typical thermal resistance | per diode | $R_{\theta JC}$ | 2.8 | °C/W |
| | per device | | 1.5 | |
| | per device | $R_{\theta JA}$ (1)(2) | 58 | |

**Notes**

(1) The heat generated must be less than the thermal conductivity from junction-to-ambient: $dP_D/dT_J < 1/R_{\theta JA}$ - junction-to -mount

(2) Free air, without heatsink

| ORDERING INFORMATION (Example) | | | | |
|---|---|---|---|---|
| PREFERRED P/N | UNIT WEIGHT (g) | PACKAGE CODE | BASE QUANTITY | DELIVERY MODE |
| V20D202C-M3/I | 0.55 | I | 2000/reel | 13" diameter plastic tape and reel |
| V20D202CHM3_B/I (1) | 0.55 | I | 2000/reel | 13" diameter plastic tape and reel |

**Note**

(1) AEC-Q101 qualified

## RATINGS AND CHARACTERISTICS CURVES ($T_A$ = 25 °C unless otherwise noted)

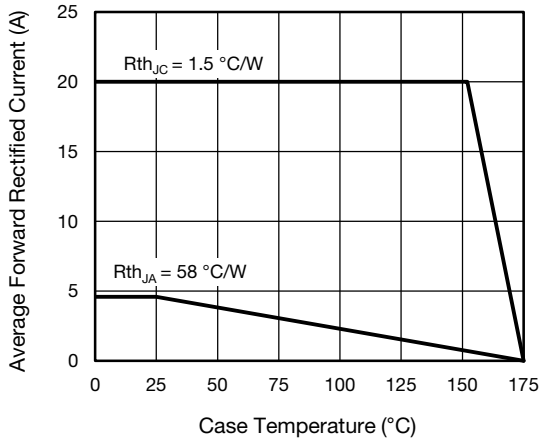



Fig. 1 - Forward Current Derating Curve

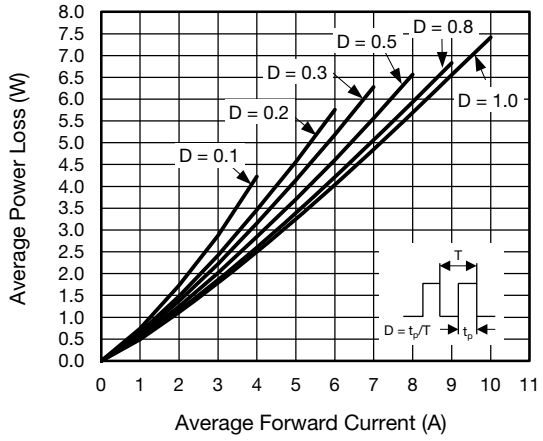



Fig. 2 - Forward Power Loss Characteristics



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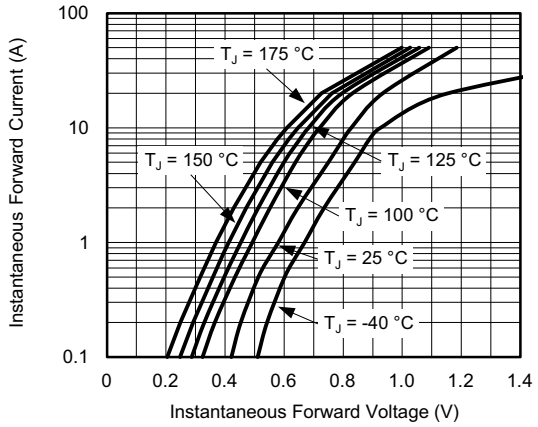



Fig. 3 - Typical Instantaneous Forward Characteristics

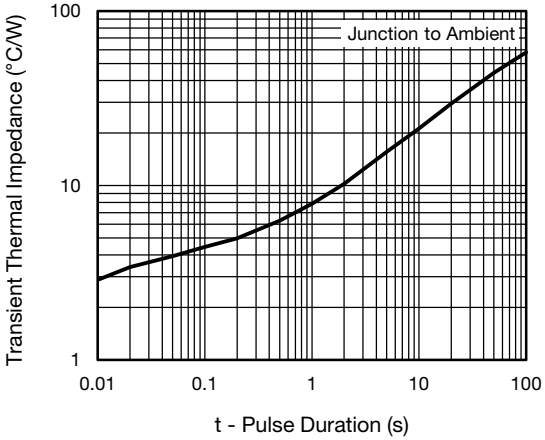



Fig. 6 - Typical Transient Thermal Impedance

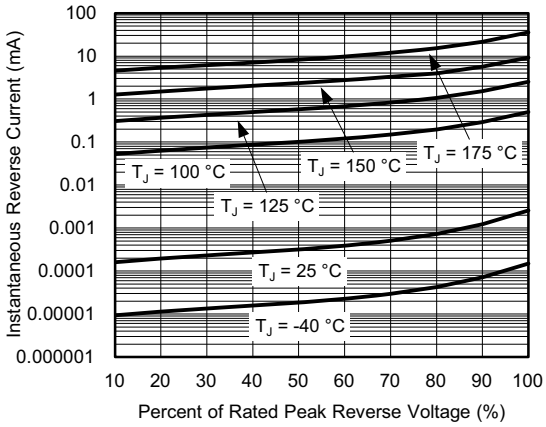



Fig. 4 - Typical Reverse Characteristics

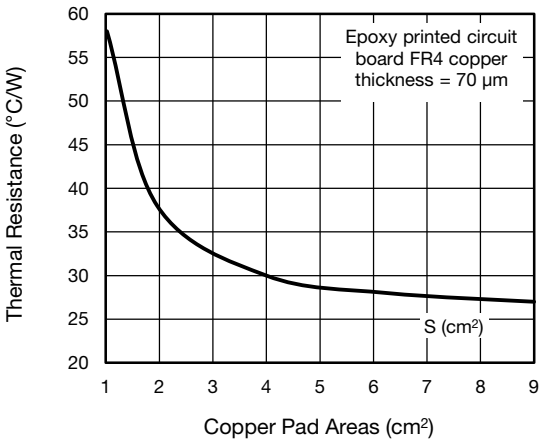



Fig. 7 - Thermal Resistance Junction-to-Ambient vs. Copper Pad Areas

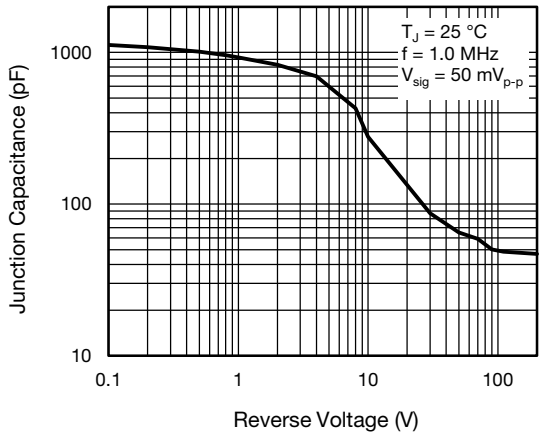
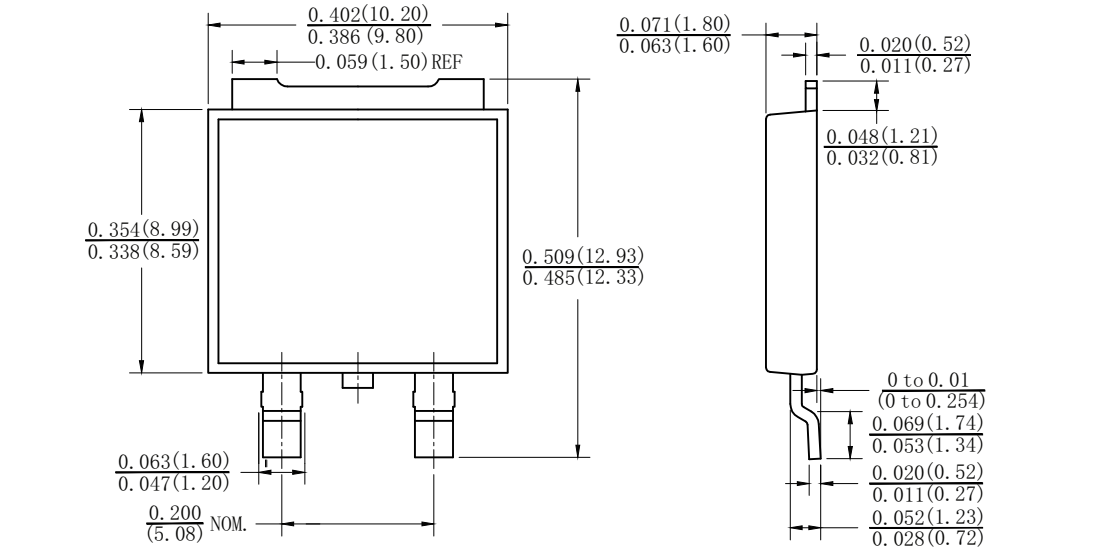



Fig. 5 - Typical Junction Capacitance

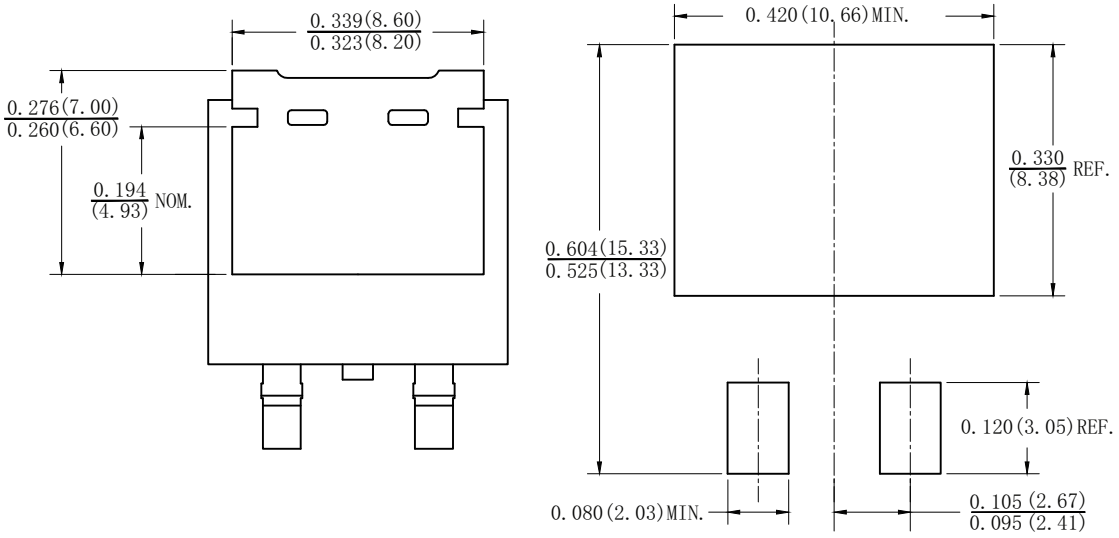

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**PACKAGE OUTLINE DIMENSIONS** in inches (millimeters)

SMPD (TO-263AC)




Mounting Pad Layout





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